

TIPCALC — Tip and total

TI-84 Evo / Evo-T · Python App
2026-07-26

Enter a bill amount and a tip percentage; the calculator returns the tip and the grand total, both rounded to two decimals.

WHAT IT ASKS YOU

1. Bill amount: the pre-tip total, e.g. 48.50
2. Tip percent: the percentage as a whole number, e.g. 18 (not 0.18)

WHAT IT SHOWS

- Tip: the tip on its own, 2 decimals
- Total: bill + tip, 2 decimals

THE MATH

```
tip    = bill * pct / 100
total  = bill + tip
```

GET IT ONTO THE CALCULATOR

1. Open **connectevo.ti.com** in Chrome (v143+) and accept the terms.
2. Plug the calculator in with a USB-C *data* cable and switch it on.
3. Click **CONNECT TO CALCULATOR**, pick your calculator, click **Connect**.
4. Click **SEND TO CALCULATOR** and choose TIPCALC.py.
5. Check the name reads TIPCALC, not PYTHON01. Send.

RUN IT

`apps` → **Python App** → highlight TIPCALC → **Run**. Answer each prompt and press `enter`.

CHECK IT WORKS

```
TIP CALCULATOR
Bill amount: 48.50
Tip percent: 18
Tip: 8.73
Total: 57.23
```

GOOD TO KNOW

- Any prompt accepts an expression, not just a decimal: type 40+8.50 to split a bill, or 6*pi where that makes sense.
- Enter the tip as 18, not 0.18. The program divides by 100 for you.
- Type anything that is not a number and it prints "Not a number" and asks again - nothing crashes.
- A zero bill is valid and returns 0.00 / 0.00.
- Numeric output only, no CAS: permitted for the TI-84 family under the ACT calculator policy. Note that European exam mode disables user programs while it is active.

Mode does not matter. TIPCALC only does arithmetic, so the DEGREE / RADIAN setting has no effect on it. (For reference: Python's math module is always in radians regardless of the MODE screen, which governs TI-BASIC only.)

SOURCE CODE — TIPCALC.PY

```
# TIPCALC -- tip amount and total from a bill and a tip
percentage
# TI-84 Evo / Evo-T / TI-84 Plus CE Python / TI-83
Premium CE Python
#
# Numeric output only. No CAS. Mode does not matter --
arithmetic only.

from math import *

def getnum(msg):
    """Prompt until a number is entered.

    Accepts a plain number (48.50), a decimal comma
    (48,50), arithmetic
    (40+8.50), or an expression using math names (6*pi).
    """
    while True:
        s = input(msg)
        # European keypads and habits produce "48,50".
        # Without this line
        # eval() would read it as the tuple (48, 50) and
        # reject it.
        if "," in s and "(" not in s:
            s = s.replace(",", ".")
        try:
            return float(eval(s))
        except:
            print("Not a number")

print("TIP CALCULATOR")

bill = getnum("Bill amount: ")
pct = getnum("Tip percent: ")

tip = bill * pct / 100
total = bill + tip

print("Tip: {:.2f}".format(tip))
print("Total: {:.2f}".format(total))
```